

EgoPathBench: Evaluating Zero-Shot Egocentric Waypoint Decision-Making in Vision-Language Models

Anonymous submission

Abstract

Zero-shot waypoint navigation requires vision-language models to select, from the current first-person observation, a sequence of spatial actions that is feasible for the agent and reaches the goal, placing joint demands on the integrated spatial intelligence of today’s foundation VLMs. Existing spatial-intelligence benchmarks primarily evaluate isolated judgments of relations, directions, or targets and therefore do not directly measure the integrated navigation ability required to combine target recognition, action-consequence assessment, distance estimation, and path planning. To fill this evaluation gap, we introduce EgoPathBench, a dataset and five-task benchmark for first-person waypoint decision-making. Each question presents an egocentric RGB image, a natural-language goal, and numbered visible waypoints; a model returns traversable candidates or an ordered route. Predictions are evaluated for candidate feasibility, adjacent-edge legality, and goal arrival under point-agent or embodied geometry. EgoPathBench contains 31,852 training, 1,345 validation, and 1,111 benchmark questions and retains at least one geometrically verified reference route for every route question. Across nine VLMs, the highest EgoPath Score is only 28.3. The top-ranked model reaches 35.9% success on Point Path, but only 2.9% and 4.0% on Embodied Path and Intent Path, respectively, showing that current models remain limited in forming complete, goal-consistent routes under embodiment constraints. Beyond the evaluation data, we release the corresponding training resource. Fine-tuning Qwen 3.5 4B on the released training split raises its EgoPath Score from 3.9 to 38.9 and improves all four reported evaluations across three external spatial benchmarks, with gains of 1.4–9.6 points.

Introduction

Foundation vision-language models are increasingly used in embodied navigation to translate first-person observations and task goals into spatial actions. Recent work has explored VLMs or LLMs for explicit navigation reasoning, generalist navigation, visual next-step planning, and zero-shot Object-Nav (Zhou, Hong, and Wu 2024; Zheng et al. 2024; Zhang et al. 2024; Cai et al. 2025). Navigation decisions require more than recognizing objects or answering an individual

spatial-relation question: a model must interpret the current environment, the intended target, and the available actions to make a goal-directed spatial choice. The integrated spatial intelligence of foundation VLMs is therefore an important basis for zero-shot navigation to generalize across new scenes and goals.

Existing evaluations characterize complementary parts of spatial and embodied reasoning. SpatialVLM and Spatial-Eval evaluate metric and relational spatial judgments; VSI-Bench studies spatial understanding and memory from video observations; and 3DSRBench evaluates reasoning about 3D structure (Chen et al. 2024; Wang et al. 2024; Yang et al. 2025; Ma et al. 2025). EmbSpatial-Bench, EgoThink, and OpenEQA extend evaluation to first-person or embodied observations (Du et al. 2024; Cheng et al. 2024; Majumdar et al. 2024). These benchmarks provide useful measurements of component abilities. Navigation evaluations provide a complementary view: NavBench studies navigation comprehension and sequential execution (Qiao et al. 2025a), while outcomes in full navigation systems may additionally depend on mapping, localization, memory, control, replanning, and recovery. NaviTrace evaluates two-dimensional navigation traces from a single real-world image against expert demonstrations using a semantic-aware score (Windecker et al. 2025). EgoPathBench complements these directions by isolating waypoint selection from the surrounding navigation system and directly evaluating whether a foundation VLM can choose a geometrically feasible, goal-consistent route over currently visible actions.

To fill this evaluation gap, we introduce EgoPathBench, a dataset and five-task benchmark for first-person waypoint decision-making. EgoPathBench uses waypoint selection in the current view as a controlled measurement substrate for integrated spatial intelligence. Each question presents an egocentric RGB image, a natural-language prompt, and numbered visible waypoints; the model returns a JSON list containing traversable candidates or an ordered route. Across five tasks, the benchmark organizes target recognition and grounding, waypoint-to-scene correspondence, geometric action consequences, distance and route efficiency, embodied feasibility, and multi-step route planning within a unified action-selection interface. The resulting construct tests whether these abilities jointly support a first-person decision with explicit action consequences. Figure 1 illustrates this

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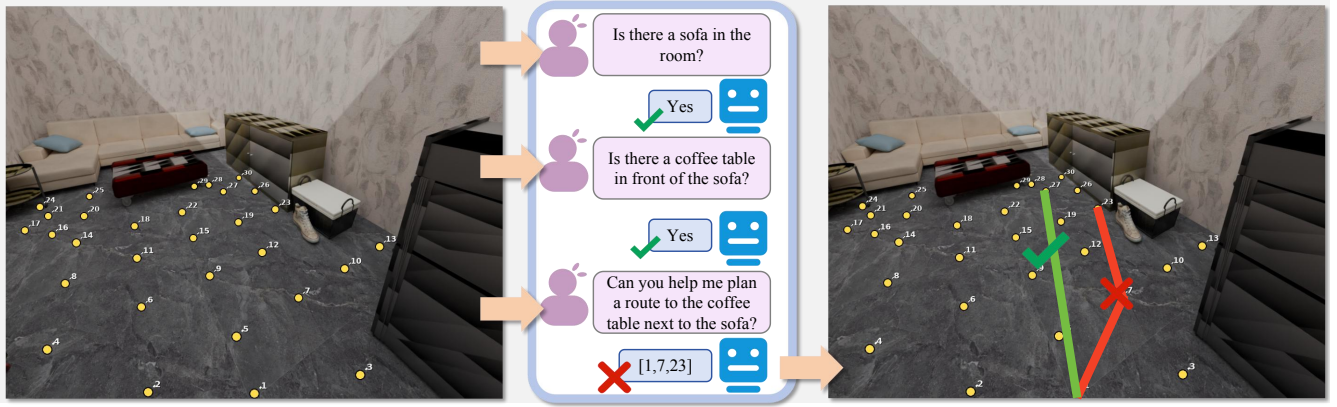


Figure 1: Conceptual motivation for EgoPathBench. On the same first-person scene, familiar recognition and local spatial questions can be answered correctly. The right panel contrasts a complete, goal-reaching route with an incorrect route that violates the route and goal criteria. This contrast motivates evaluating integrated, embodiment-aware route decisions rather than component visual questions alone.

motivation conceptually: the same scene can support familiar recognition and local spatial judgments, while forming a complete, legal, and goal-consistent waypoint route requires these abilities to operate jointly over an action sequence.

Point Traversability and Embodied Traversability ask which visible waypoints are feasible for a point agent or an embodied agent. Explicit Point-Goal Path and Explicit Embodied-Goal Path share the scene, target, and candidate space while asking the model to route the two agent types to an explicit target. Intent-Grounded Embodied Path instead describes the target through an intent and a visual cue, requiring target grounding together with embodied route selection. Point-agent and embodied tasks use the same observation and waypoint vocabulary, while the corresponding geometric feasibility graph defines the consequences of each selected action.

The ordered waypoint route makes these abilities jointly testable in a structured output. A model must select an appropriate immediate action, maintain agent-specific feasibility across every adjacent edge, and terminate at an acceptable goal. EgoPathBench therefore checks candidates, the required start, adjacent edges, the endpoint, and post-success route efficiency separately. A current-view waypoint route provides a verifiable joint spatial-decision output and a local planning representation that can be used within waypoint-based zero-shot navigation systems. Evaluating this representation through a shared observation and action interface enables direct comparison of foundation-VLM spatial decisions independently of other navigation modules.

To align the model observation, displayed actions, and their consequences, we anchor every waypoint to a scene location and construct point-agent and embodied feasibility graphs from scene geometry. Targets, visible waypoints, legal edges, acceptable endpoints, and reference routes are fixed before prompt generation. Every route question retains at least one reference route that passes the construction and geometry checks and projects into the current view. The model receives only the RGB image with numbered waypoints and

a natural-language prompt, while its prediction is evaluated using candidate feasibility, adjacent-edge legality, and goal arrival defined in the same scene.

EgoPathBench contains 31,852 training, 1,345 validation, and 1,111 benchmark questions. In addition to the benchmark split for unified model comparison, we release geometry-grounded training answers, legal reference routes, and Spatial CoT supervision, supporting both evaluation and the study of learned first-person spatial decision-making. We evaluate nine foundation VLMs, analyze where their waypoint routes fail, and examine the benchmark’s dependence on its paired visual interface and scene-grounded evaluator. We also train a Qwen 3.5 4B model to test whether the released supervision transfers to improved performance on EgoPathBench and external spatial tasks.

The results reveal a substantial gap in current models’ integrated spatial decision-making. The highest EgoPath Score across nine VLMs is 28.3. The top-ranked model reaches 35.9% success on Point Path, but only 2.9% and 4.0% on Embodied Path and Intent Path, respectively, showing a marked decline when a route must jointly satisfy embodiment and goal conditions. Fine-tuning Qwen 3.5 4B on the released training split raises its EgoPath Score from 3.9 to 38.9 and improves all four reported evaluations across three external spatial benchmarks, with gains of 1.4–9.6 points. These results show that EgoPathBench differentiates current models’ first-person spatial decisions and that the released data provides effective supervision for this ability.

The contributions are:

- We introduce EgoPathBench, which uses five first-person waypoint-decision tasks to evaluate the integrated spatial intelligence of foundation VLMs through controlled, action-valued outputs.
- We construct and release 31,852 training, 1,345 validation, and 1,111 benchmark questions by aligning rendered observations, displayed waypoints, and resolved targets with point-agent and embodied feasibility graphs, together with legal reference routes and Spatial CoT su-

pervision.

- We systematically evaluate nine foundation VLMs, identify a major capability gap in embodied route decisions, and evaluate the released training resource on EgoPathBench and three external spatial benchmarks.

Related Work

Foundation-model navigation and evaluation. R2R, REVERIE, RxR, and VLN-CE established vision-language navigation in discrete and continuous indoor environments (Anderson et al. 2018; Qi et al. 2020; Ku et al. 2020; Krantz et al. 2020). Foundation-model agents such as NavGPT, Nav-iLLM, and NaVid subsequently used language reasoning, generalist embodied modeling, and video-based planning for navigation (Zhou, Hong, and Wu 2024; Zheng et al. 2024; Zhang et al. 2024). Recent zero-shot systems organize spatial decisions through different interfaces. SmartWay combines waypoint prediction with history-aware backtracking; VLFM, InstructNav, and CA-Nav use occupancy or value maps and sub-instruction constraints; AgenticNav and P2DNav expose pixel-level or hierarchical direction-to-grounding actions; and DreamNav predicts trajectories rather than isolated points (Shi et al. 2025; Yokoyama et al. 2024; Long et al. 2024; Chen et al. 2025; Li et al. 2026; Sheng et al. 2026; Wang et al. 2025). Open-Nav studies spatio-temporal reasoning with open-source models, Nav-R1 learns structured navigation traces, and LHPR-VLN extends evaluation to decision consistency across long-horizon subtasks (Qiao et al. 2025b; Liu et al. 2025; Song et al. 2025). Across full navigation systems, outcomes can depend on different combinations of candidate generation, mapping, memory, control, and replanning over repeated observations. EgoPathBench measures a controlled capability within this process: given the current first-person observation and a shared vocabulary of visible spatial actions, can a foundation VLM form a geometrically feasible and goal-consistent waypoint decision?

NaviTrace is the closest evaluation to our setting. It presents a single real RGB image, a navigation instruction, and an embodiment description, and asks a VLM to produce a continuous trace in image space (Windecker et al. 2025). The prediction is scored by Dynamic Time Warping against human-annotated traces, endpoint error, and embodiment-conditioned penalties derived from pixel semantics. NaviTrace therefore measures agreement with expert navigation demonstrations under a score that combines trace similarity, endpoint accuracy, and embodiment-conditioned semantic penalties.

EgoPathBench differs in the supervision associated with each observation. Each image is registered to an underlying 3D scene representation, from which we derive visible waypoints, agent-specific feasibility, direct traversability between waypoints, and acceptable goal regions. This scene backing allows an arbitrary predicted route to be evaluated by the geometric outcomes of its selected actions: whether its waypoints are feasible, whether every consecutive transition satisfies the specified agent constraints, and whether the route reaches the goal. A reference route certifies that a geometrically valid solution exists, but is not the unique

trajectory that a prediction must imitate. Thus, NaviTrace evaluates trajectory agreement defined by expert demonstrations and image semantics, whereas EgoPathBench evaluates waypoint-action outcomes defined by scene geometry.

Spatial intelligence and embodied reasoning. Existing spatial-intelligence benchmarks examine complementary components of foundation VLMs’ spatial ability. SpatialVLM and SpatialEval emphasize distance, direction, and spatial relations; VSI-Bench studies video-based spatial understanding and memory; ViewSpatial evaluates multi-perspective spatial localization; and 3DSRBench focuses on 3D structure (Chen et al. 2024; Wang et al. 2024; Yang et al. 2025; Li et al. 2025a; Ma et al. 2025). EmbSpatial-Bench, EgoThink, and OpenEQA introduce first-person or embodied observations (Du et al. 2024; Cheng et al. 2024; Majumdar et al. 2024). Embodied3DBench further evaluates low-level embodied skills including grounding, affordance, and trajectory prediction; CapNav studies capability-conditioned navigation under agent-specific mobility constraints; and IndustryNav evaluates active planning and collision-aware navigation in dynamic industrial environments (Zhang et al. 2026; Su et al. 2026; Li et al. 2025b). Together, these benchmarks show that embodied spatial intelligence involves not only recognizing scene relations, but also determining where an agent can act and what spatial consequences its actions produce.

EgoPathBench organizes these abilities into a joint decision with an explicit action interpretation. A model must relate the target, candidate locations, agent constraints, and route structure in the current observation, then return either a traversable waypoint set or an ordered route. The prediction is evaluated through geometry-defined candidate feasibility, consecutive-edge legality, and goal attainment. EgoPathBench thereby connects low-level embodied spatial cues to ordered first-person waypoint decisions with explicit scene-grounded consequences.

EgoPathBench Dataset and Benchmark

Benchmark Formulation

EgoPathBench represents navigation decisions as selections over a visible waypoint vocabulary. Each example provides an egocentric RGB image with numbered candidate locations and a natural-language task. The model returns a JSON list of display IDs. For the two traversability tasks, the list denotes an unordered set of traversable candidates. For the three path tasks, it denotes an ordered route from a specified start to an acceptable goal.

Formally, an example consists of an egocentric observation o , a displayed waypoint set A , a target specification t , and a task-specific feasibility graph G . Every display ID in A is anchored to a 3D scene location. Given o , A , and t , the model predicts either a subset of A or an ordered sequence over A . Traversability tasks evaluate the feasibility of selected vertices. Route tasks additionally require the specified start, legal consecutive edges in G , and an endpoint in the acceptable goal set.

Point and Embodied tasks share the same observation and waypoint vocabulary but use different feasibility graphs. The

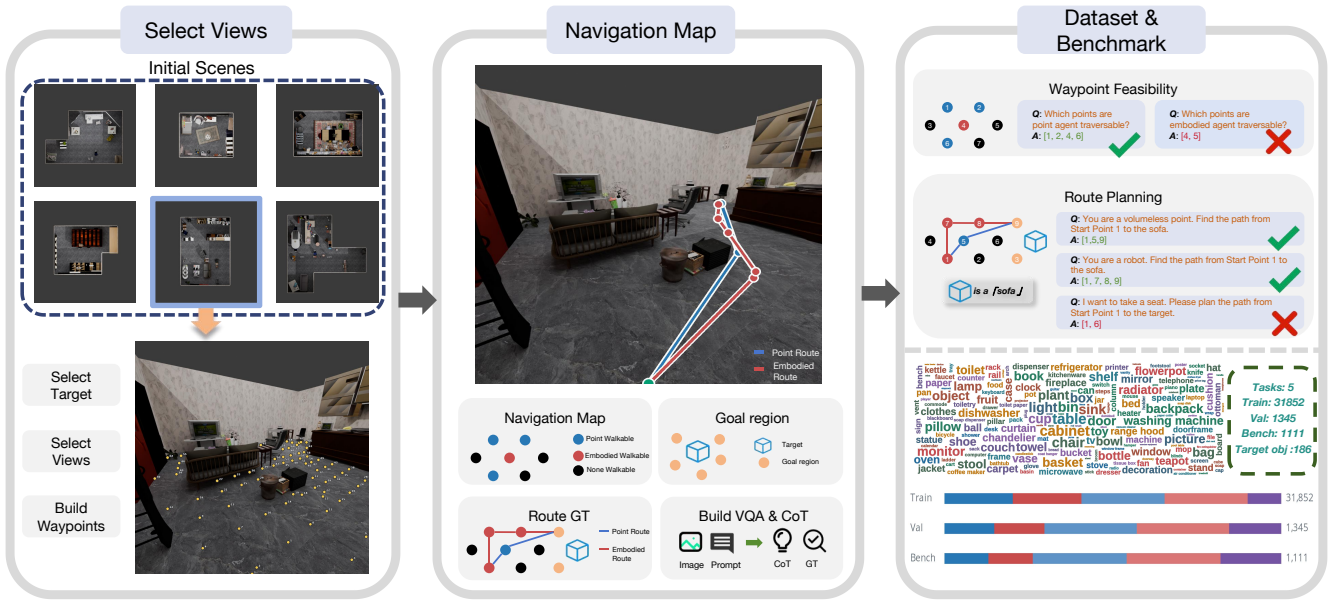


Figure 2: EgoPathBench construction pipeline. We select scenes, first-person views, and targets, then build visible waypoints, point- and embodiment-specific navigation maps, goal regions, and geometry-verified reference routes. These fixed scene annotations support image-question and Spatial CoT generation, yielding two waypoint-feasibility tasks and three route-planning tasks across the training, validation, and benchmark splits.

point graph captures geometric connectivity without body width, whereas the embodied graph additionally accounts for the agent footprint. The same visually selected action can therefore have different consequences under the two agent models.

EgoPathBench measures joint waypoint decision making from limited first-person visual evidence. The model must identify the target, associate displayed IDs with scene locations, and select actions that jointly satisfy traversability, embodiment, route continuity, and goal-reaching requirements. The model acts from the current observation, while the consequences of its choices are evaluated against the registered scene geometry.

Five-Task Design

EgoPathBench contains two traversability tasks and three route tasks. The five tasks progressively introduce target grounding, embodiment, and route-composition requirements while retaining a common input and output interface.

Point Traversability asks the model to select all candidates traversable by a point agent. **Embodied Traversability** uses the same scene view and candidate space but accounts for the agent footprint, requiring sufficient clearance at the selected locations. This paired design tests how embodiment changes action feasibility within the same observation.

Point Path provides an explicit target and asks for an ordered point-agent route. **Embodied Path** holds the target, view, and candidate space fixed but evaluates the route under embodied feasibility. **Intent Path** replaces the explicit object name with an intent and a visual cue. It therefore requires the model to resolve the intended object before selecting an embodied route.

Data Construction and Annotation

Figure 2 summarizes the construction process. We first establish scene provenance and first-person scene-view records, then generate visible waypoints and geometric task labels. Natural-language questions and Spatial CoT are produced only after the target, action space, and reference route have been fixed. Quality filtering and challenge-set selection produce the final release.

Scenes, views, and targets. We use unified simulatable indoor assets normalized through InternScenes (Zhong et al. 2025). The scenes retained in our release derive from 3RScan (Wald et al. 2019), ScanNet (Dai et al. 2017), ARKitScenes (Baruch et al. 2021), and Matterport3D (Chang et al. 2017). We use these assets for scene geometry and rendering, and construct our own first-person observations, waypoint anno-

Task	Goal	Agent	Output	Scored constraints	N
Point Trav.	—	Point	Set	Traversability	146
Embodied Trav.	—	Emb.	Set	Feasibility	146
Point Path	Explicit	Point	Route	Edges, endpoint	309
Embodied Path	Explicit	Emb.	Route	Footprint, edges, goal	309
Intent Path	Intent	Emb.	Route	Intent, footprint, edges	201

Table 1: Five EgoPathBench tasks. The shared first-person waypoint interface is specialized by goal specification, agent geometry, output structure, evaluated constraints, and benchmark question count.

tations, route labels, and question data.

For each scene, we sample first-person cameras in traversable space and remove views whose immediate camera neighborhood is obstructed. Target instances are then filtered by frustum projection, observation distance, projected size, and visible surface evidence, excluding targets that are outside the image, too small, or heavily occluded. Shared camera parameters and scene geometry bind each retained target and RGB observation to a common scene-view record, which provides the spatial basis for subsequent waypoint and route annotation.

Waypoints and geometric labels. For each retained view, we project two types of markers into the image: ground action candidates and selected visible object- or structural-surface locations used as non-traversable negatives. Depth, ray-visibility, and marker-spacing checks associate every displayed ID with a distinct scene location while reducing marker overlap and foreground occlusion. Route sheets use visible ground waypoints and explicitly insert the waypoints required by retained reference routes.

We evaluate ground actions under both point and embodied geometry. Traversability examples therefore mix feasible ground actions, clearance-sensitive ground actions, and visible surface negatives, so the task cannot be solved by selecting every displayed ID. Point and Embodied Traversability share the same view, while their labels follow the corresponding agent-feasibility conditions. Each displayed candidate stores its ID, image projection, and scene coordinate; the numbered RGB overlay and the geometric annotations therefore refer to the same action vocabulary.

Paired route construction. For each route unit, we anchor the start on visible floor near the bottom of the image and generate feasible goal locations around the target footprint. We then search separately through point and embodied free space, convert each continuous path into sparse route waypoints, and project the required waypoints back into the current image.

Construction checks route connectivity, task-specific feasibility, arrival in the target region, and the availability of every required waypoint through the displayed action interface. Routes that fail these checks are discarded. Point Path and Embodied Path consequently share the target, start, and scene view while retaining routes validated for their respective agent geometries. Every retained route question has at least one scene-verified reference route to its target.

Question text and Spatial CoT. Target identity, displayed waypoints, goal region, and reference-route identity are fixed before language generation. Explicit-target tasks name the required object directly. For Intent Path, we construct a view-specific candidate universe from visible objects and generate a target description using supported relations, attributes, colors, or distance cues.

The resulting description is resolved again against the objects available in the current view. We retain it only when it uniquely identifies the fixed target without unsupported cues. Intent Path is derived from an accepted Embodied Path instance, so changing the linguistic specification does not change its target or route geometry.

The training split additionally includes Spatial CoT. GPT-

5.5 verbalizes the fixed target, candidates, feasibility labels, legal edges, and reference route into task-specific reasoning text. The exported answer is checked against the formal annotation, keeping language generation downstream of the geometric ground truth.

Quality Control

Quality control applies geometry and view-action checks before language is attached. Candidate locations, route edges, and target regions share one scene coordinate system; disconnected or colliding routes, routes that miss the target region, and examples without required display waypoints are removed. The RGB image, waypoint overlay, and ID-to-waypoint record come from the same rendered view, and point and embodied routes are validated under their corresponding agent geometries.

Prompts may describe only targets and scene facts fixed during construction. Explicit-target questions are checked against the selected target, while intent questions must resolve uniquely among objects visible from the current view. A full visual-language alignment audit then checks targets, prompts, displayed waypoints, reference routes, and answers across the formal release. Ambiguous or unsupported prompts are repaired or removed without changing the established geometric labels or route identities.

Benchmark Selection and Release

After constructing the full question pool, we select the formal benchmark as a challenging subset emphasizing scene clutter, traversability boundaries, point-embodied feasibility differences, competing targets, and route composition.

Selection operates on complete task bundles. Point and Embodied Traversability form a paired view bundle; Point Path and Embodied Path form a paired route bundle with a shared target and candidate space; eligible Intent Path questions accompany their corresponding embodied routes. Bundle-level selection prevents the benchmark from retaining only one side of a paired comparison.

We then select scenes while balancing task coverage, geometric difficulty, and scene diversity and limiting concentration within InternScenes. Splits are formed by source group, so the same original scan or related regions do not cross training, validation, and benchmark sets. The final release contains 31,852 training questions, 1,345 validation questions, and 1,111 benchmark questions across the five first-person waypoint-decision tasks.

Experiments

Experimental Setup

The benchmark contains 1,111 questions: 146 Point Traversability, 146 Embodied Traversability, 309 Point Path, 309 Embodied Path, and 201 Intent Path. Traversability uses candidate-level balanced accuracy (BA), the mean recall across the task-feasible and infeasible classes, and F1, the precision-recall harmonic mean for feasible candidates. For route tasks, valid path rate (VPR) measures complete route legality, success rate (SR) additionally requires an acceptable endpoint, and success weighted by path length (SPL)

Model	Score	Point Trav.		Embodied Trav.		Point Path			Embodied Path			Intent Path		
		BA	F1	BA	F1	VPR	SR	SPL	VPR	SR	SPL	VPR	SR	SPL
Zero-shot foundation VLMs														
Gemini 3.1 Pro	28.3	76.5	79.0	72.7	56.2	63.7	35.9	28.7	9.1	2.9	2.4	10.4	4.0	3.3
GPT-5.5	27.3	74.1	77.1	77.3	62.5	60.5	31.1	24.8	5.5	1.3	1.3	9.0	1.5	1.4
Claude Opus 4.8	25.6	77.9	74.3	73.5	59.6	66.0	21.0	16.7	12.0	1.6	1.5	14.4	2.5	2.2
MiniMax M3	21.8	77.0	76.2	68.7	52.8	50.8	13.3	8.7	5.5	1.9	1.7	9.0	2.5	2.3
Qwen 3.6	16.4	67.4	72.1	64.8	49.1	49.2	15.2	10.9	2.6	1.0	0.9	5.0	1.5	1.3
Mistral L3	15.8	65.2	70.8	68.5	52.5	35.6	11.0	5.8	0.3	0.0	0.0	3.5	0.5	0.5
Llama 4	14.9	66.8	66.5	66.0	49.8	36.2	8.7	5.9	1.9	0.0	0.0	2.0	0.0	0.0
Kimi K2.6	9.7	60.2	69.8	57.8	44.2	40.5	11.7	8.2	1.3	0.7	0.5	1.5	0.5	0.4
Grok 4.3	1.4	52.3	58.4	50.0	35.8	18.4	2.6	1.2	5.2	0.0	0.0	3.5	0.0	0.0
Training-resource study														
Qwen3.5-4B base	3.9	54.6	55.6	54.9	33.1	9.1	0.7	0.1	0.3	0.0	0.0	0.0	0.0	0.0
+ EgoPathBench SFT	38.9	89.3	89.2	83.4	71.2	77.0	31.4	28.6	34.9	7.1	6.9	44.8	10.4	10.0

Table 2: EgoPathBench leaderboard (%). The upper block compares zero-shot foundation VLMs; the lower block compares Qwen3.5-4B before and after training on EgoPathBench. Score is the equal-weight five-task macro-average. BA/F1 evaluate traversability, while VPR, SR, and SPL evaluate route validity, goal-reaching success, and efficiency.

discounts successful routes that are longer than the shortest legal reference while assigning zero to failures. EgoPath Score equally averages chance-adjusted traversability BA and route SR across the five tasks. Formal definitions are provided in the supplementary material.

We evaluate nine foundation VLMs. Every model receives the same image, task prompt, visible waypoint IDs, and JSON output protocol, and every prediction is scored by the same evaluator. Endpoint identifiers, generation settings, training hyperparameters, and control-specific protocols are provided in the supplementary material.

Zero-Shot VLM Results

Table 2 gives the complete zero-shot leaderboard. Gemini 3.1 Pro ranks first with an EgoPath Score of 28.3, followed by GPT-5.5 at 27.3 and Claude Opus 4.8 at 25.6.

Candidate-level traversability is markedly stronger than complete route construction. The best Point and Embodied Traversability BA values are 77.9% and 77.3%, with best F1 scores of 79.0% and 62.5%. For Point Path, the strongest VPR is 66.0%, but SR falls to 35.9% and SPL to 28.7%. Thus, plausible local actions do not by themselves produce a legal, goal-reaching route.

Tasks with embodiment constraints produce the largest drop. On Embodied Path, the best VPR, SR, and SPL are 12.0%, 2.9%, and 2.4%; on Intent Path, they are 14.4%, 4.0%, and 3.3%. The VPR–SR gap indicates complementary failures in edge legality and endpoint selection, while low

SPL largely reflects scarce complete successes.

Figure 4 places the zero-shot leaderboard against a same-question human reference under the identical task interface. The human profile is higher on all five task axes and reaches an EgoPath Score of 54.2, compared with 28.6 for the strongest VLM on these questions. The human advantage appears in both candidate-feasibility judgments and all three route tasks, indicating that the gap is not driven by any single task or metric but reflects a broader limitation in first-person spatial decision-making.

Where Complete Routes Fail

Figure 3 decomposes route predictions into five parallel diagnostics, ordered from output validity and target-endpoint selection to the legality of the initial action, the complete route, and their joint success. Across tasks, 96.3–96.8% of outputs are evaluable routes, so the output protocol is not the main bottleneck. Goal-consistent endpoint rates are much lower at 28.9% for Point Path, 4.8% for Embodied Path, and 5.2% for Intent Path, exposing substantial difficulty in grounding the requested target to a terminal waypoint. A model may recognize where the target object is yet fail to identify a nearby terminal waypoint that is feasible for the specified agent.

Initial action selection is considerably stronger: 96.0%, 90.4%, and 88.0% of predictions take a legal first edge. This local feasibility does not carry through the selected sequence. Full-route legality falls to 46.8% for Point Path,

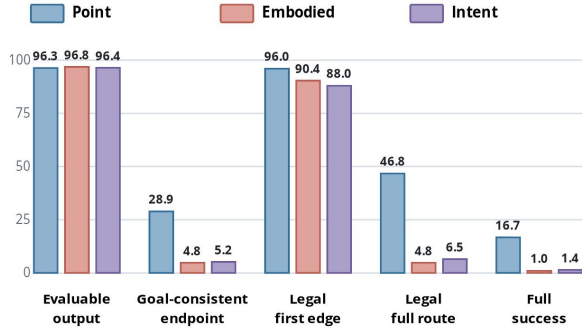


Figure 3: Route diagnostics pooled over nine VLMs (%). Rates over all predictions measure evaluable output, goal-consistent endpoint selection, legal first action, full-route legality, and joint success.

4.8% for Embodied Path, and 6.5% for Intent Path; among predictions with a legal first edge, 51.3%, 94.7%, and 92.7% contain an illegal later edge. Joint success, which requires both complete-route legality and a goal-consistent endpoint, is only 16.7%, 1.0%, and 1.4%. The diagnostics therefore expose two distinct limitations: selecting the intended terminal waypoint and maintaining agent-specific feasibility from the current position to that endpoint.

This suffix failure persists after controlling for both ends of the decision. Among predictions with a legal first edge and acceptable goal, 42.2% of Point, 75.8% of Embodied, and 69.4% of Intent routes still contain an illegal intermediate edge. On the half of questions with fewer displayed waypoints, rates remain similar at 42.8%, 76.6%, and 67.3%, so the pattern is not confined to dense action overlays. Illegal-edge incidence is already 90.6% and 88.4% on one-edge-reference Embodied and Intent questions, rising to 94.0% and 92.4% for references with at least three edges. The central signal is whether the intermediate sequence remains valid under agent geometry; additional model-level and length-stratified results appear in the supplementary material.

Through experiments, we find that removing the image or mismatching the waypoint overlay degrades route performance, showing that predictions depend on the paired visual input rather than the prompt alone. Geometry perturbations preserve the released conclusions. The supplementary material reports the complete protocols, control results, human-reference details, and scene-consequence audit.

Training-Resource Evaluation

We fine-tune Qwen3.5-4B on the released EgoPathBench training split. The base model and selected checkpoint use the same EgoPathBench contract and evaluation items from VSI-Bench Route Planning, SpatialEval-VTQA, and 3DSRBench (Yang et al. 2025; Wang et al. 2024; Ma et al. 2025). This evaluates the released training data in- and out-of-domain.

The lower block of Table 2 reports the in-domain comparison. The fine-tuned checkpoint improves every reported metric and reaches an EgoPath Score of 38.9.

The selected checkpoint improves all reported external

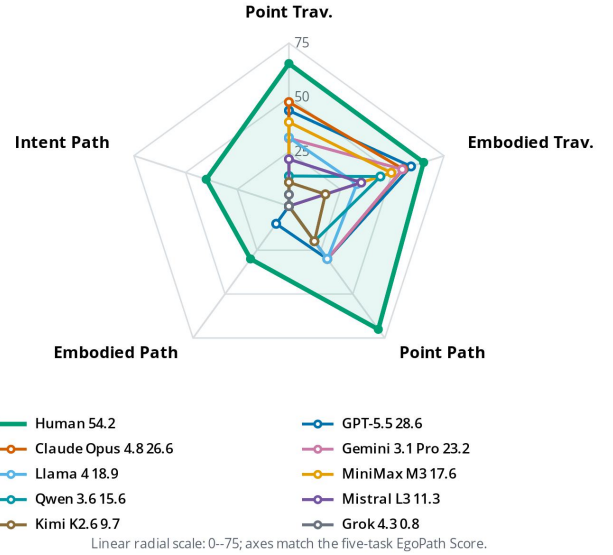


Figure 4: Same-question comparison of humans and nine VLMs across five tasks, with traversability measured by 2BA – 1, route performance by success rate, and EgoPath Score shown in the legend.

Benchmark	Setting	Base	SFT	Δ
VSI-Bench Route Planning	Full	29.38	33.51	+4.13
	Debiased	20.18	24.56	+4.38
SpatialEval-VTQA	Full	61.8	71.4	+9.6
3DSRBench	Full	58.0	59.4	+1.4

Table 3: Performance of Qwen3.5-4B on three spatial benchmarks before and after supervised fine-tuning (SFT) on EgoPathBench (%).

evaluations. On VSI-Bench Route Planning, performance rises from 29.38% to 33.51% in the Full setting (+4.13) and from 20.18% to 24.56% in the Debiased setting (+4.38). Under the Full settings, accuracy also increases by 9.6 points on SpatialEval-VTQA and 1.4 points on 3DSRBench, showing transfer to spatial tasks outside EgoPathBench.

Conclusion

EgoPathBench evaluates whether VLMs turn first-person observations and goals into feasible, goal-reaching actions across five scene-grounded tasks spanning feasibility, grounding, route legality, goal arrival, and efficiency.

Across nine VLMs, the best EgoPath Score is 28.3; the leader reaches 35.9% Point Path success but only 2.9% and 4.0% on Embodied and Intent Path. Fine-tuning improves Qwen 3.5 4B from 3.9 to 38.9 and yields gains on all four reported evaluations across three external spatial benchmarks, supporting both evaluation and training. The gap between locally plausible actions and complete embodied routes identifies sustained geometric and goal consistency across multi-step decisions as a central direction for future VLM research.

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